Simulation: Batch distillation

BATCH DISTILLATION 1 CALCULATION RESULTS

Operation Step 1:

STREAM PROPERTIES

| Name | Pot Charge | Accumulator | Pot Residue | Distillate |
|---|---|---|---|---|
| - - Overall - - | | | | |
| Molar flow kmol | 71.5527 | 41.2935 | 100.0092 | 0.1841 |
| Mass flow kg | 3097.6099 | 2479.3572 | 6066.2036 | 10.3529 |
| Temp C | 80.2389 | 82.4446 | 142.0314 | 81.2891 |
| Pres bar | 1.0130 | 1.0130 | 1.0130 | 1.0130 |
| Vapor mole fraction | 0.000 | 1.105E-05 | 0.000 | 0.000 |
| Enth MJ | -21268. | -12685. | -25329. | -56.153 |
| Tc C | 257.5353 | 235.2141 | 440.3150 | 239.0848 |
| Pc bar | 58.6826 | 47.7569 | 117.6872 | 53.9094 |
| Std. sp gr. wtr = 1 | 0.822 | 0.794 | 1.087 | 0.799 |
| Std. sp gr. air = 1 | 1.495 | 2.073 | 2.094 | 1.942 |
| Degree API | 40.6544 | 46.7489 | -1.3678 | 45.6866 |
| Average mol wt | 43.2913 | 60.0423 | 60.6564 | 56.2480 |
| Actual dens kg/m3 | 756.8923 | 608.4261 | 964.8813 | 729.2065 |
| Actual vol m3 | 4.0925 | 4.0750 | 5.9978 | 0.0142 |
| Std liq m3 | 3.7687 | 3.1233 | 5.5789 | 0.0130 |
| Std vap 0 C m3 | 1603.7571 | 925.5381 | 2241.5715 | 4.1254 |
| - - Vapor only - - | | | | |
| Molar flow kmol | | 0.0227 | | |
| Mass flow kg | | 1.3630 | | |
| Average mol wt | | 60.0083 | | |
| Actual dens kg/m3 | | 2.1118 | | |
| Actual vol m3 | | 0.6454 | | |
| Std liq m3 | | 0.0017 | | |
| Std vap 0 C m3 | | 0.5091 | | |
| Cp kJ/kg-K | | 1.7026 | | |
| Z factor | | 0.9736 | | |
| Visc N-s/m2 | | 9.301e-06 | | |
| Th cond W/m-K | | 0.0224 | | |
| - - Liquid only - - | | | | |
| Molar flow kmol | 71.5527 | 41.2708 | 100.0092 | 0.1841 |
| Mass flow kg | 3097.6099 | 2477.9939 | 6066.2036 | 10.3529 |
| Average mol wt | 43.2913 | 60.0423 | 60.6564 | 56.2480 |
| Actual dens kg/m3 | 756.8923 | 722.5254 | 964.8813 | 729.2065 |
| Actual vol m3 | 4.0925 | 3.4296 | 6.2870 | 0.0142 |
| Std liq m3 | 3.7687 | 3.1216 | 5.5789 | 0.0130 |
| Std vap 0 C m3 | 1603.7571 | 925.0291 | 2241.5715 | 4.1254 |
| Cp kJ/kg-K | 3.5054 | 3.3993 | 2.3643 | 3.4065 |
| Z factor | 0.0024 | 0.0033 | 0.0026 | 0.0031 |
| Visc N-s/m2 | 0.0004483 | 0.0005015 | 0.0003773 | 0.0004961 |
| Th cond W/m-K | 0.1506 | 0.1219 | 0.1335 | 0.1261 |
| Surf. tens. N/m | 0.0202 | 0.0164 | 0.0287 | 0.0170 |

Component mole fractions

| | | | | |
|---|---|---|---|---|
| Isopropanol | 0.600658 | 0.998707 | 0.017385 | 0.908548 |
| Water | 0.399342 | 0.001281 | 0.285512 | 0.091445 |
| DiMth Sulfoxide | 0.000000 | 0.000012 | 0.697103 | 0.000007 |

Operation Step 2:

STREAM PROPERTIES

| Name | Accumulator | Pot Residue | Distillate |
|---|---|---|---|
| - - Overall - - | | | |
| Molar flow kmol | 8.7748 | 91.2344 | 0.1966 |
| Mass flow kg | 231.0943 | 5835.1094 | 3.5674 |
| Temp C | 81.6701 | 155.1982 | 98.5217 |
| Pres bar | 1.0130 | 1.0130 | 1.0130 |
| Vapor mole fraction | 0.01001 | 0.000 | 0.000 |
| Enth MJ | -2510.9 | -22693. | -55.098 |
| Tc C | 310.1375 | 446.5912 | 372.7422 |
| Pc bar | 100.4756 | 106.9352 | 217.9982 |
| Std. sp gr. wtr = 1 | 0.895 | 1.097 | 0.997 |
| Std. sp gr. air = 1 | 0.909 | 2.208 | 0.626 |
| Degree API | 26.5887 | -2.4750 | 10.3711 |
| Average mol wt | 26.3362 | 63.9573 | 18.1432 |
| Actual dens kg/m3 | 83.0079 | 959.8750 | 955.9047 |
| Actual vol m3 | 2.7840 | 5.8501 | 0.0037 |
| Std liq m3 | 0.2582 | 5.3207 | 0.0036 |
| Std vap 0 C m3 | 196.6749 | 2044.8966 | 4.4070 |
| - - Vapor only - - | | | |
| Molar flow kmol | 0.0879 | 0.0001 | |
| Mass flow kg | 3.5833 | 0.0039 | |
| Average mol wt | 40.7794 | 33.9610 | |
| Actual dens kg/m3 | 1.4254 | 0.9757 | |
| Actual vol m3 | 2.5139 | 0.0040 | |
| Std liq m3 | 0.0043 | 0.0000 | |
| Std vap 0 C m3 | 1.9695 | 0.0026 | |
| Cp kJ/kg-K | 1.7368 | 1.6254 | |
| Z factor | 0.9824 | 0.9900 | |
| Visc N-s/m2 | 1.015e-05 | 1.244e-05 | |
| Th cond W/m-K | 0.0217 | 0.0232 | |
| - - Liquid only - - | | | |
| Molar flow kmol | 8.6869 | 91.2343 | 0.1966 |
| Mass flow kg | 227.5110 | 5835.1055 | 3.5674 |
| Average mol wt | 26.1901 | 63.9573 | 18.1432 |
| Actual dens kg/m3 | 842.3177 | 960.5107 | 955.9048 |
| Actual vol m3 | 0.2701 | 6.0750 | 0.0037 |
| Std liq m3 | 0.2539 | 5.3207 | 0.0036 |
| Std vap 0 C m3 | 194.7054 | 2044.8940 | 4.4070 |
| Cp kJ/kg-K | 3.8367 | 2.3332 | 4.2114 |
| Z factor | 0.0013 | 0.0027 | 0.0008 |
| Visc N-s/m2 | 0.0003738 | 0.0003747 | 0.0002862 |
| Th cond W/m-K | 0.2905 | 0.1246 | 0.6637 |
| Surf. tens. N/m | 0.0329 | 0.0273 | 0.0580 |

Component mole fractions

| | | | |
|---|---|---|---|
| Isopropanol | 0.197742 | 0.000039 | 0.003047 |
| Water | 0.802258 | 0.235812 | 0.996953 |
| DiMth Sulfoxide | 0.000000 | 0.764150 | 0.000000 |

Operation Step 3:

STREAM PROPERTIES

| Name | Accumulator | Pot Residue | Distillate |
|---|---|---|---|
| - - Overall - - | | | |
| Molar flow kmol | 19.5114 | 71.7231 | 0.1574 |
| Mass flow kg | 352.7416 | 5482.3677 | 3.5762 |
| Temp C | 99.8998 | 183.8788 | 101.5942 |
| Pres bar | 1.0130 | 1.0130 | 1.0130 |
| Vapor mole fraction | 0.000 | 0.000 | 0.000 |
| Enth MJ | -5463.6 | -16957. | -43.691 |
| Tc C | 374.3753 | 452.2162 | 392.4992 |
| Pc bar | 221.3230 | 62.0678 | 237.2394 |
| Std. sp gr. wtr = 1 | 1.000 | 1.104 | 1.026 |
| Std. sp gr. air = 1 | 0.624 | 2.639 | 0.784 |
| Degree API | 9.9681 | -3.2756 | 6.4059 |
| Average mol wt | 18.0788 | 76.4380 | 22.7152 |
| Actual dens kg/m3 | 958.4564 | 931.7993 | 973.3488 |
| Actual vol m3 | 0.3680 | 5.7836 | 0.0037 |
| Std liq m3 | 0.3527 | 4.9680 | 0.0035 |
| Std vap 0 C m3 | 437.3209 | 1607.5757 | 3.5287 |
| - - Vapor only - - | | | |
| Molar flow kmol | | 0.0001 | |
| Mass flow kg | | 0.0079 | |
| Average mol wt | | 68.6628 | |
| Actual dens kg/m3 | | 1.8680 | |
| Actual vol m3 | | 0.0042 | |
| Std liq m3 | | 0.0000 | |
| Std vap 0 C m3 | | 0.0026 | |
| Cp kJ/kg-K | | 1.5138 | |
| Z factor | | 0.9799 | |
| Visc N-s/m2 | | 1.087e-05 | |
| Th cond W/m-K | | 0.0207 | |
| - - Liquid only - - | | | |
| Molar flow kmol | 19.5114 | 71.7230 | 0.1574 |
| Mass flow kg | 352.7416 | 5482.3594 | 3.5762 |
| Average mol wt | 18.0788 | 76.4380 | 22.7152 |
| Actual dens kg/m3 | 958.4564 | 932.4653 | 973.3488 |
| Actual vol m3 | 0.3680 | 5.8794 | 0.0037 |
| Std liq m3 | 0.3527 | 4.9680 | 0.0035 |
| Std vap 0 C m3 | 437.3209 | 1607.5729 | 3.5287 |
| Cp kJ/kg-K | 4.2100 | 2.3162 | 3.6354 |
| Z factor | 0.0008 | 0.0031 | 0.0011 |
| Visc N-s/m2 | 0.0002821 | 0.0004125 | 0.0002971 |
| Th cond W/m-K | 0.6729 | 0.1025 | 0.4664 |
| Surf. tens. N/m | 0.0585 | 0.0226 | 0.0508 |

| Component mole fractions | | | |
|---|---|---|---|
| Isopropanol | 0.000180 | 0.000000 | 0.000000 |
| Water | 0.998885 | 0.028227 | 0.921819 |
| DiMth Sulfoxide | 0.000934 | 0.971773 | 0.078181 |

Operation Step 4:

STREAM PROPERTIES

| Name | Accumulator | Pot Residue | Distillate |
|---|---|---|---|
| - - Overall - - | | | |
| Molar flow kmol | 17.5626 | 54.1605 | 0.3886 |
| Mass flow kg | 1252.0337 | 4230.3335 | 30.1852 |
| Temp C | 171.1395 | 188.7203 | 187.4501 |
| Pres bar | 1.0130 | 1.0130 | 1.0130 |
| Vapor mole fraction | 0.008973 | 0.000 | 0.000 |
| Enth MJ | -4240.6 | -12700. | -91.317 |
| Tc C | 450.1214 | 452.8400 | 452.6825 |
| Pc bar | 79.7714 | 56.5894 | 57.9808 |
| Std. sp gr. wtr = 1 | 1.101 | 1.104 | 1.104 |
| Std. sp gr. air = 1 | 2.461 | 2.697 | 2.682 |
| Degree API | -2.9800 | -3.3631 | -3.3411 |
| Average mol wt | 71.2897 | 78.1074 | 77.6796 |
| Actual dens kg/m3 | 179.2986 | 896.0194 | 928.7385 |
| Actual vol m3 | 6.9830 | 4.7213 | 0.0325 |
| Std liq m3 | 1.1372 | 3.8308 | 0.0273 |
| Std vap 0 C m3 | 393.6422 | 1213.9333 | 8.7096 |
| - - Vapor only - - | | | |
| Molar flow kmol | 0.1576 | 0.0043 | |
| Mass flow kg | 7.8036 | 0.3373 | |
| Average mol wt | 49.5159 | 77.9652 | |
| Actual dens kg/m3 | 1.3771 | 2.1062 | |
| Actual vol m3 | 5.6669 | 0.1601 | |
| Std liq m3 | 0.0072 | 0.0003 | |
| Std vap 0 C m3 | 3.5324 | 0.0970 | |
| Cp kJ/kg-K | 1.5496 | 1.5052 | |
| Z factor | 0.9861 | 0.9764 | |
| Visc N-s/m2 | 1.164e-05 | 1.061e-05 | |
| Th cond W/m-K | 0.0216 | 0.0206 | |
| - - Liquid only - - | | | |
| Molar flow kmol | 17.4050 | 54.1561 | 0.3886 |
| Mass flow kg | 1244.2301 | 4230.0000 | 30.1852 |
| Average mol wt | 71.4869 | 78.1074 | 77.6796 |
| Actual dens kg/m3 | 945.3953 | 927.4034 | 928.7385 |
| Actual vol m3 | 1.3161 | 4.5611 | 0.0325 |
| Std liq m3 | 1.1300 | 3.8305 | 0.0273 |
| Std vap 0 C m3 | 390.1099 | 1213.8363 | 8.7096 |
| Cp kJ/kg-K | 2.3094 | 2.3231 | 2.3211 |
| Z factor | 0.0029 | 0.0031 | 0.0031 |
| Visc N-s/m2 | 0.0003955 | 0.0004191 | 0.0004174 |
| Th cond W/m-K | 0.1107 | 0.0997 | 0.1004 |
| Surf. tens. N/m | 0.0246 | 0.0220 | 0.0221 |

| Component mole fractions | | | |
|---|---|---|---|
| Isopropanol | 0.000000 | 0.000000 | 0.000000 |
| Water | 0.113860 | 0.000459 | 0.007575 |
| DiMth Sulfoxide | 0.886140 | 0.999541 | 0.992425 |